Yujin Choi

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Research Positions

Sep. 2026 – Present	**Assistant Professor**, Sejong University *Department of Computer Science and Engineering*
Apr. 2026 – Aug. 2026	**Postdoctoral Researcher**, Nanyang Technological University, Singapore *The College of Computing and Data Science* Host: Prof. Jaehong Yoon
Mar. 2026 – Aug. 2026	**Postdoctoral Researcher**, Ulsan National Institute of Science and Technology *Department of Industrial Engineering* Mentor: Prof. Saerom Park

Education

Mar. 2021 – Feb. 2026	**Ph.D.** in Industrial Engineering, **Seoul National University**, Seoul, Korea Dissertation: *Trustworthy Generative Models: from Privacy to Fairness* Advisor: Prof. Jaewook Lee, GPA: 3.84 / 4.3
Mar. 2017 – Feb. 2021	**B.Eng.** in Industrial Engineering & **B.Sc.** in Mathematics (Double Major) **Yonsei University**, Seoul, Korea GPA: 3.91 / 4.3
Mar. 2014 – Feb. 2017	**Sejong Science High School**, Seoul, Korea

Publications

(†: equal contribution.; *: corresponding author.; underline = 1st/corr. author.)

Journals

[J9] H. Kim, J. Park, W. Lee, J. Lee, & **Y. Choi***. Exploring the Effect of Multi-Step Ascent in Sharpness-Aware Minimization. *IEEE Access.* IF: 3.6, top 35.0%

[J8] **Y. Choi,** J. Park, Y. Park, J. Lee, & J. Byun*. Differentially Private Upsampling for Enhanced Anomaly Detection in Imbalanced Data. *Engineering Applications of Artificial Intelligence (2026)*. IF: 8.0, top 2.8%

[J7] **Y. Choi,** D. Kim, & J. Lee*. Temporal Consistency Ensemble Empirical Mode Decomposition for Forecasting Practical Metal Price. *Engineering Applications of Artificial Intelligence, 158, 111490 (2025)*. IF: 8.0, top 2.8%

[J6] J. Byun, **Y. Choi,** & J. Lee*. Improving the Utility of Differentially Private Clustering through Dynamical Processing. *Pattern Recognition, 157, 110890 (2025)*. IF: 7.6, top 9.0%

[J5] J. Byun, **Y. Choi,** J. Lee, & S. Park*. Privacy-Preserving Inference Resistant to Model Extraction Attacks. *Expert Systems with Applications, 256, 124830 (2024)*. IF: 7.5, top 6.6%

[J4] J. Park, H. Kim, **Y. Choi,** W. Lee, & J. Lee*. Fast Sharpness-Aware Training for Periodic Time Series Classification and Forecasting. *Applied Soft Computing, 144, 110467 (2023)*. IF: 6.6, top 13.7%

[J3] J. Byun, S. Park, **Y. Choi,** & J. Lee*. Efficient Homomorphic Encryption Framework for Privacy-Preserving Regression. *Applied Intelligence, 53(9), 10114–10129 (2023)*. IF: 3.5, top 41.2%

[J2] **Y. Choi,** J. Park, J. Lee, & H. Kim*. Exploring Diverse Feature Extractions for Adversarial Audio Detection. *IEEE Access, 11, 2351–2360 (2023)*. IF: 3.6, top 35.0%

[J1] J. Park, **Y. Choi,** J. Byun, J. Lee, & S. Park*. Efficient Differentially Private Kernel Support Vector Classifier for Multi-Class Classification. *Information Sciences, 619, 889–907 (2023)*. IF: 6.8, top 8.1%

Top Conferences

[C9] W. Jeong†, **Y. Choi†,** D. Kim, S. Park & J. Lee*. Leveraging Pathology Co-occurrence for Test-Time Adaptation in Chest X-Ray Diagnosis, *Medical Image Computing and Computer Assisted Intervention (MICCAI)* 2026, *accepted*

[C8] H. Kim, J. Park, **Y. Choi,** & J. Lee*. Stability Analysis of Sharpness-Aware Minimization. *International Conference on Machine Learning (ICML) 2026, accepted*

[C7] J. Park, S. Lee, W. Jeong, **Y. Choi,** & J. Lee*. Leveraging Priors via Diffusion Bridge for Time Series Generation. *SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2026*

[C6] J. Park, **Y. Choi**, & J. Lee*. Multi-Class Support Vector Machine with Differential Privacy. *Advances in Neural Information Processing Systems (NeurIPS) 2025*

[C5] **Y. Choi**†, Y. Park†, J. Byun, J. Lee, & J. Park*. Safeguarding Privacy of Retrieval Data against Membership Inference Attacks: Is This Query Too Close to Home?. *Findings of EMNLP 2025, Suzhou, China*

[C4] J. Park†, **Y. Choi**†, & J. Lee*. In-Distribution Public Data Synthesis with Diffusion Models for Differentially Private Image Classification. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 12236–12246 (2024)*

[C3] **Y. Choi**†, J. Park†, H. Kim, J. Lee, & S. Park*. Fair Sampling in Diffusion Models through Switching Mechanism. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 38(20), 21995–22003 (2024)*

[C2] H. Kim, J. Park, **Y. Choi,** & J. Lee*. Fantastic Robustness Measures: The Secrets of Robust Generalization. *Advances in Neural Information Processing Systems (NeurIPS), 36, 48793–48818 (2023)*

[C1] J. Park, H. Kim, **Y. Choi,** & J. Lee*. Differentially Private Sharpness-Aware Training. *International Conference on Machine Learning (ICML), 27204–27224 (2023)*

Patents

2025.05.28	Clustering Method and Apparatus Supporting Differential Privacy Reg. No. 10-2815746 | App. No. 1020220185186, filed 2022.12.27
2023.06.07	Apparatus and Method for Learning Differential Privacy Multi-Classification Based on Kernel Support Vector App. No. 10-2023-0073178
2023.07.05	Method and System for Detecting Adversarial Speech Example App. No. 1020230087024

Preprints & Under Review

[7] **Y. Choi,** & J. Yoon*. Safe Few-Step Generation via Velocity Editing, Submitted, *NeurIPS 2026*

[6] Y. Hee, **Y. Choi**, & J. Yoon*. Confidence-Aware Tool Orchestration for Robust Video Understanding, Submitted, *NeurIPS 2026*

[5] D.Kim†, G.Shin†, **Y. Choi,** S.Park, & J.Lee*. A Locally Tokenized Generative Model for Robust Time-Series Watermarking, Submitted, *NeurIPS 2026*

[4] **Y. Choi,** J. Park, Y. Park, & J. Lee*. Multiple Hyperplanes with Dual Encoding Support Vector Machine for Industrial Anomaly Detection. *Minor revision, IEEE Transactions on Industrial Informatics* (IF: 9.9)

[3] **Y. Choi,** J. Park, J. Byun, & J. Lee*. Leveraging Programmatically Generated Synthetic Data for Differentially Private Diffusion Training, *submitted, Information Science* (IF: 6.8)

[2] J. Park, **Y. Choi,** Y. Park, & J. Lee*. Privacy-Preserving Asymmetric Banach Kernel Encoding for Support Vector Classification. *Submitted to Pattern Recognition* (IF: 7.6)

[1] H. Kim, J. Park, **Y. Choi,** S. Lee, & J. Lee*. BayesNAM: Leveraging Inconsistency for Reliable Explanations. *1st revision, Pattern Recognition* (IF: 7.6)

Scholarships and Fellowships

2025	Youlchon AI Scholarship, Youlchon AI Star
2023–2025	Doctoral Student Research Grant, Ministry of Science and ICT, Korea (Graduate Research Fellowship in Science and Engineering)
2019–2020	National Science and Technology Scholarship (full tuition), Ministry of Science, Korea

Teaching Experience

Lecturer

Spring 2025	Financial Machine Learning, Soongsil University, Seoul, Korea

Fall 2024	Financial Programming 2, Soongsil University, Seoul, Korea

Teaching Assistant (TA), Industrial Engineering (Prof. Jaewook Lee), SNU

2022	Mathematical Methods for Industrial Management, Seoul National University

TA & Instructor, Corporate Education Programs, SNU

2023–2024	HD Hyundai: Nonlinear Time Series Analysis
2022	Woori Bank: Big Data Analysis with Python and Deep Learning with PyTorch
2022–2024	SNU Big Data FinTech: Optimization and Machine Learning
2024–2025	SNU Big Data AI CEO Program
2021–2025	Samsung Data Scientist for Device Solution: Optimization
2021–2022	Samsung Data Scientist for Device Solution: Linear Algebra
	Samsung Best TA Award, 2021

References

Jaewook Lee	Professor of Industrial Engineering, Seoul National University jaewook@snu.ac.kr
Saerom Park	Assistant professor of Industrial Engineering, UNIST srompark@unist.ac.kr
Jaehong Yoon	Assistant professor of Computer and Data Science, Nanyang Technological University jaehong.yoon@ntu.edu.sg
Hoki Kim	Assistant professor of Industrial Security, Chung-Ang University hokikim@cau.ac.kr